随机过程

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Chapter 5. Renewal processes

§ 5.1 Introduction

§ 5.2 The strong law of large numbers and convergence WP1

§ 5.3 Strong law for renewal processes

§ 5.4 Renewal–reward processes; time averages

§ 5.1 Introduction

回顾以下概念。

设概率空间 $(\Omega, \mathcal{F}, P)$, 其中 Ω 是样本空间，$\mathcal{F}$ 是事件类，而 P 是概率。

概率是从事件类到实数的一个映射，满足

- 规范性：$P(\Omega) = 1$.
- 非负性：$P(A) \geq 0$ for all $A \in \mathcal{F}$.
- 可列可加性：设 $A_i \in \mathcal{F}, i \geq 0$ 是一列两两不相交的事件，则 $P(\bigcup_{i\geq 0} A_i) = \sum_{i\geq 0} P(A_i)$.

§ 5.1 Introduction

随机变量是从样本空间到实数轴的映射，即

$$X : \Omega \to \mathbb{R}$$

任意 $\omega \in \Omega$, 有唯一的 $X(\omega) \in \mathbb{R}$ 与之对应，满足对于任意 x,

$\{X \leq x\} \in \mathcal{F}$(是一个事件)。

多维随机变量是从样本空间到多维实空间的映射，即

$$X : \Omega \to \mathbb{R}^n$$

§ 5.1 Introduction

一维随机变量可以用累积分布函数

$$F_X(x) = P\{X \leq x\}, x \in \mathbb{R}$$

来刻画。

n 维随机变量可以用联合累积分布函数

$$F_{X_1,X_2,\cdots,X_n}(x_1, x_2, \cdots, x_n) = P\{X_1 \leq x_1, X_2 \leq x_2, \cdots, X_n \leq x_n\}, x_1^n \in \mathbb{R}^n$$

来刻画。

§ 5.1 Introduction

思考题：

随机变量序列是怎样的映射？

随机过程是怎样的映射？

二者什么关系？如何刻画它们？

§ 5.1 Introduction

随机变量序列是从样本空间到数列空间的一个映射；

$X_n(\omega), n \in \mathbb{Z}, \omega \in \Omega$;

随机过程是从样本空间到函数空间的一个映射；$X(t;\omega), t \in \mathbb{R}, \omega \in \Omega$;

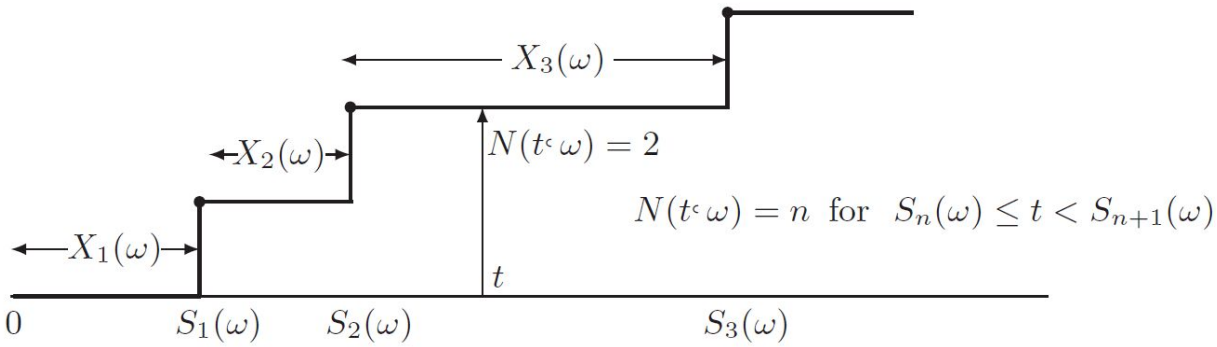
对于随机过程而言，需要用任意有限维的联合分布函数来描述，即，对于任意 n, 任意 $-\infty < t_1 < t_2 < \cdots < t_n < \infty$, 我们需要掌握 $(X(t_1), X(t_2), \cdots, X(t_n))$ 的联合累积分布函数。这对于一般化的随机过程是不可能的。

思考题：你认为比较简单的随机过程是什么？

§ 5.1 Introduction

Recall that an arrival process can be specified by either one of the joint distributions

- of the arrival epochs, $S_1, S_2, \cdots, S_n, \cdots$
- of the interarrival times, $X_1, X_2, \cdots, X_n, \cdots$
- of the counting random variables, $N(t), t \geq 0$.

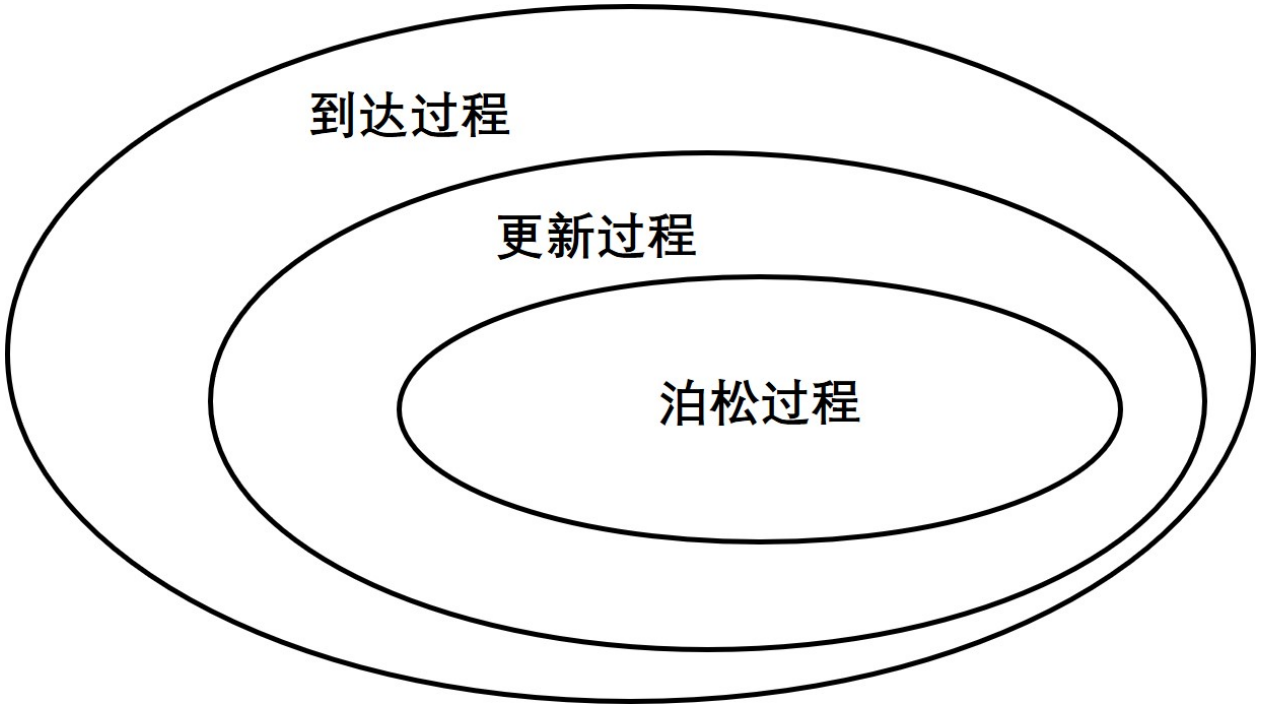


- $X_i = S_i - S_{i-1}$ for $i \geqslant 2$ and $X_1 = S_1$; $S_n = \sum_{i=1}^{n} X_i$.
- $\{S_n \leqslant t\} = \{N(t) \geqslant n\}$ for all $n \geqslant 1,\ t > 0$.

§ 5.1 Introduction

Definition 1

A renewal process is an arrival process for which the sequence of interarrival times $X_1, X_2, \ldots$ are i.i.d..



§ 5.1 Introduction

Example 1

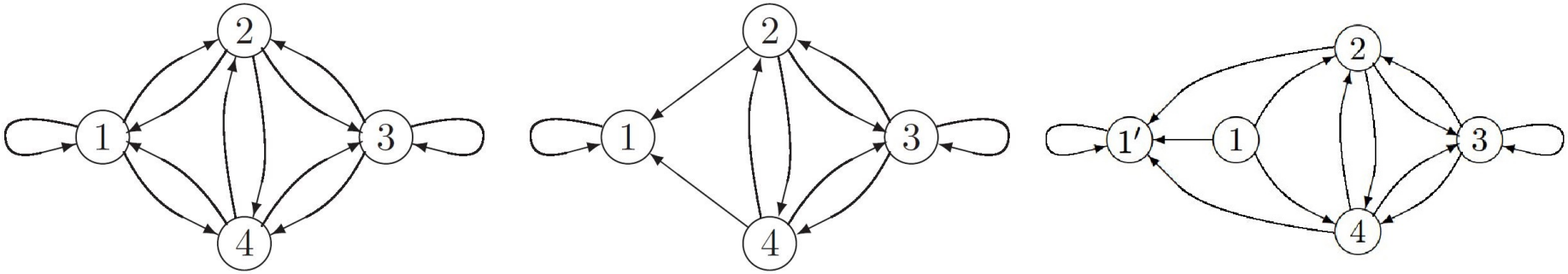
Markov chains, for any given recurrent state, have renewals on successive visits to that state.
Assume that a recurrent finite-state Markov chain with transition matrix $[P]$ starts in state i at time 0. Then on the first return to state i, say at time n, the Markov chain, from time n on, is a probabilistic replica of the chain starting at time 0.
Each subsequent return to state i at a given time n starts a new probabilistic replica of the Markov chain starting in state i at time 0. Thus the sequence of entry times to state i can be viewed as the arrival epochs of a renewal process.

It is almost obvious that the intervals between visits to a given recurrent state are independent.

§ 5.1 Introduction

A non-obvious result that will arise from studying renewals is that the expected recurrence time between visits to state i is $1/\pi_i$. We could have derived that from Markov chains directly, but using renewals is cleaner.

For example, suppose the Markov chain is given by the figure. Then we can calculate the expect first-passage recurrent time from state 1 to state 1 using the trick of trapping state .



§ 5.1 Introduction

We have seen that the expect first-passage recurrent time

$$v_1 = 1 + \sum_{k\neq 1} P_{1,k} v_k,$$

where for each $j \neq 1$, v_j satisfies $v_j = 1 + \sum_{k\neq 1} P_{j,k} v_k$.
Multiplying π_j on both sides of the equations and summing up, we have

$$\begin{aligned}\sum_j \pi_j v_j =& 1 + \sum_j \sum_{k\neq 1} \pi_j P_{j,k} v_k \\ =& 1 + \sum_{k\neq 1} \sum_j \pi_j P_{j,k} v_k \\ =& 1 + \sum_{k\neq 1} \pi_k v_k.\end{aligned}$$

Thus, $\pi_1 v_1 = 1$ and $v_1 = 1/\pi_1$.
思考题：考虑有限状态 Markov 链 $X_1, X_2, \cdots, X_n, \cdots$。设其是遍历的，稳态概率（也是极限概率）分布记作 π。在 n 长的序列中，状态“1”大致发生多少次呢？

§ 5.1 Introduction

Example 2. 考虑掷一个硬币的 Bernoulli 试验，其中正面朝上（称之为成功）的概率是 p。定义随机变量 X,

$X = 1$, 如果正面朝上;

$X = 0$, 如果反面朝上

则我们可以考虑如下分布的概率质量函数、均值、方差、矩生成函数等

- Bernoulli 分布（一次试验成功的次数）
- 二项分布（n 次试验中成功的次数）
- 几何分布（首次成功需要的次数；成功后再次成功所需要的次数）
- Pascal 分布, 又名负二项分布（第 r 次成功所需要的次数）

§ 5.1 Introduction

Example 3. 以速率 $\lambda > 0$ “到达” 的泊松过程。

则我们可以考虑如下分布的概率密度函数、均值、方差、矩生成函数等

- 指数分布（“首次到达”，间隔分布）：到达间隔 X 服从指数分布 $\Pr\{X > t\} = e^{-\lambda t}$, $t \geq 0$.
- 泊松分布：$(0, t]$ 到达次数
- Erlang 分布（第 n 次到达的时刻）

§ 5.1 Introduction

Example 4. The G/G/m queue 顾客到达间隔服从一个一般分布，IID；

服务时间也服从另一个一般分布，IID，有 m 个服务窗口。

先到先服务。first-come-first-served

若都在忙，就排队。

到达的是一个到达过程 $N(t)$ 表示 $(0, t]$ 内到达的顾客数；而这个系统还嵌入一个更新过程，$N^r(t)$, $(0, t]$ 内有多少次顾客看到了空系统。

§ 5.1 Introduction

Definition 2

A renewal process is an arrival process for which the sequence of interarrival times $X_1, X_2, \ldots$ are i.i.d..

- Renewal processes are characterized by "starting over" at each arrival epoch.
- Throughout our study of renewal processes, we use $\overline{X}$ and $\mathrm{E}[X]$ interchangeably to denote the mean inter-renewal interval, and use σ_X^2 or simply σ^2 to denote the variance of the inter-renewal interval. We will usually assume that $\overline{X}$ is finite.
- Much of this chapter will be devoted to understanding the behavior of $N(t)$ and $N(t)/t$ as t becomes large.

Chapter 5. Renewal processes

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§ 5.4 Renewal–reward processes; time averages

Convergence of random variables

Recall that the four types of convergence of random variables.

Definition 3

A sequence $X_1, X_2, \ldots$ of random variables is said to **converge in distribution** to a random variable X if

$$\lim_{n\to\infty} F_n(x) = F(x), \tag{1}$$

for each $x \in \mathbb{R}$ at which F is continuous. Here F_n and F are the cumulative distribution functions of random variables X_n and X correspondingly.

Definition 4

A sequence X_n of random variables **converges in probability** towards X if for all $\varepsilon > 0$

$$\lim_{n\to\infty} P\{|X_n - X| > \varepsilon\} = 0. \tag{2}$$

Convergence of random variables

Definition 5

To say that the sequence X_n converges almost surely or almost everywhere or **with probability 1** or strongly towards X means that

$$P\{\lim_{n\to\infty} X_n = X\} = 1. \tag{3}$$

Definition 6

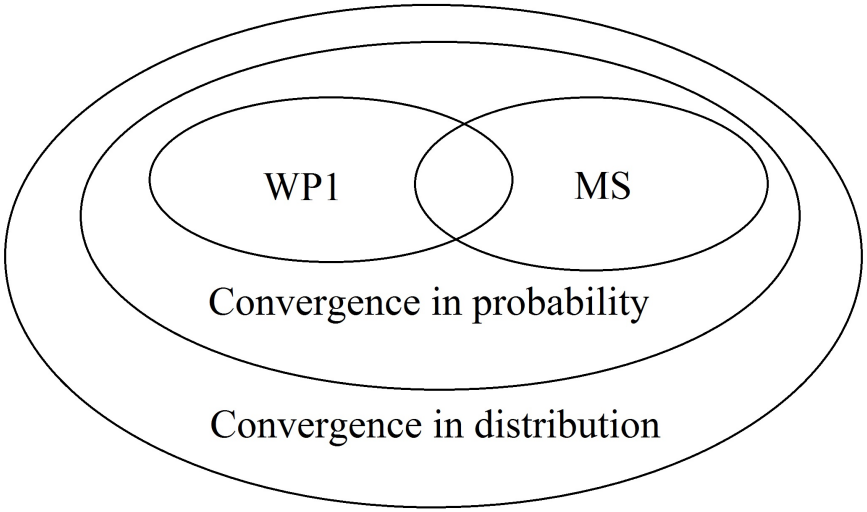
A sequence $X_1, X_2, \ldots$ of random variables is said to **converge in mean square** to X if

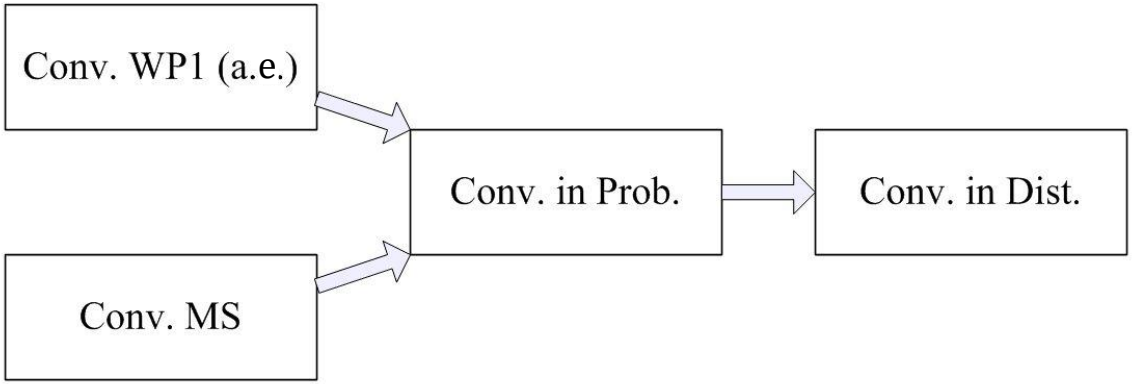
$$\lim_{n\to\infty} \mathrm{E}[(X_n - X)^2] = 0, \tag{4}$$

where the operator E denotes the expected value.

Convergence of random variables

The relationship between these forms of convergence can be shown as





Convergence of random variables

Example 2: convergence in probability but not WP1

For each $n \geqslant 1$, there exists a unique (k, i) such that

$$n = 2^k + i, i \leqslant 2^k - 1.$$

Consider a series r.v.s $\{X_n; n \geqslant 1\}$ on the sample space $\Omega = [0, 1]$, X_n is the indicator variable of the interval $[\frac{i}{2^k}, \frac{i+1}{2^k}]$.

Obviously, each sample path $\{X_n(\omega)\}$ for $\omega \in [0, 1]$ is not convergent. Therefore $\{X_n; n \geqslant 1\}$ does NOT converge WP1.
However, for each $n \geqslant 1$, and $\varepsilon < 1$,

$$P\{|X_n - 0| > \varepsilon\} = \frac{1}{2^k} < \frac{2}{n}.$$

Thus, $\lim_{n\to\infty} P\{|X_n - 0| > \varepsilon\} = 0$, i.e., $\{X_n(\omega)\}$ converges in probability.

5.2.2 Strong law of large numbers

Theorem 1 (SLLN)

For each integer $n \geqslant 1$, let $S_n = X_1 + \cdots + X_n$, where $X_1, X_2, \ldots$ are i.i.d. r.v.s satisfying $\mathrm{E}[|X|] < \infty$. Then

$$P\left\{\omega : \lim_{n\to\infty} \frac{S_n(\omega)}{n} = \overline{X}\right\} = 1.$$

The strong law is quite different from the other forms of convergence. It focusses directly on sample paths from $n = 1$ to ∞.

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§ 5.3 Strong law for renewal processes

Lemma 2

Let $\{N(t); t > 0\}$ be a renewal counting process with inter-renewal rv.s $\{X_n; n \geqslant 1\}$. Then (whether or not $\overline{X} < \infty$), $\lim_{t\to\infty} N(t) = \infty$ WP1 and $\lim_{t\to\infty} \mathrm{E}[N(t)] = \infty$.

证明思路（作业）：

（1）论证每个样本函数都有极限，或有限，或无穷。（因为样本函数是不减的）

（2）论证 $\{N(t) < n\}$ 这个事件应包含事件 $\{\lim_{t\to\infty} N(t) < n\}$

所以，$P\{\lim N(t) < n\} \leq P\{N(t) < n\} = P\{S_n > t\} \to 0$, 因为 S_n 是随机变量。

§ 5.3 Strong law for renewal processes

Lemma 3

Let $\{Z_n; n \geqslant 1\}$ be a sequence of rv.s such that $\lim_{n\to\infty} Z_n = \alpha$ WP1. Let f be a real valued function of a real variable that is continuous at α. Then

$$\lim_{n\to\infty} f(Z_n) = f(\alpha) \qquad \text{WP1.}$$

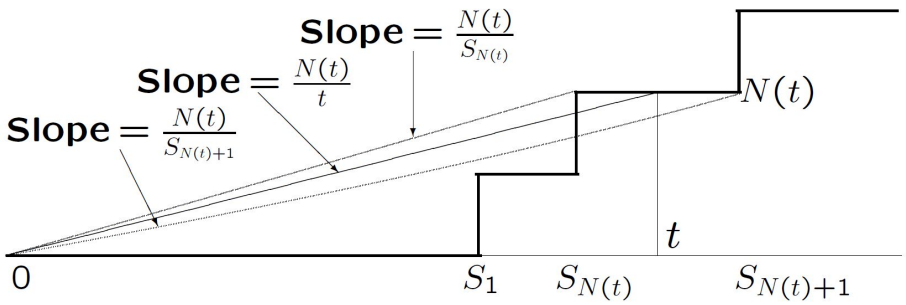
§ 5.3 Strong law for renewal processes

Theorem 2 (Strong law for renewal processes)

For a renewal process with mean inter-renewal interval $\overline{X} < \infty$,

$$\lim_{t\to\infty} N(t)/t = 1/\overline{X} \quad \text{WP1.}$$

i.e., $P\left\{\omega : \lim_{t\to\infty} N(t,\omega)/t = 1/\overline{X}\right\} = 1.$



$$\frac{N(t)}{S_{N(t)+1}} < \frac{N(t)}{t} \leqslant \frac{N(t)}{S_{N(t)}}.$$

§ 5.3 Strong law for renewal processes

Proof:

Let $n = N(t)$, then from the lemmas, we have

$$\lim_{t\to\infty} \frac{N(t)}{S_{N(t)}} = \lim_{n\to\infty} \frac{n}{S_n} = \frac{1}{\overline{X}} \quad \text{WP1.}$$

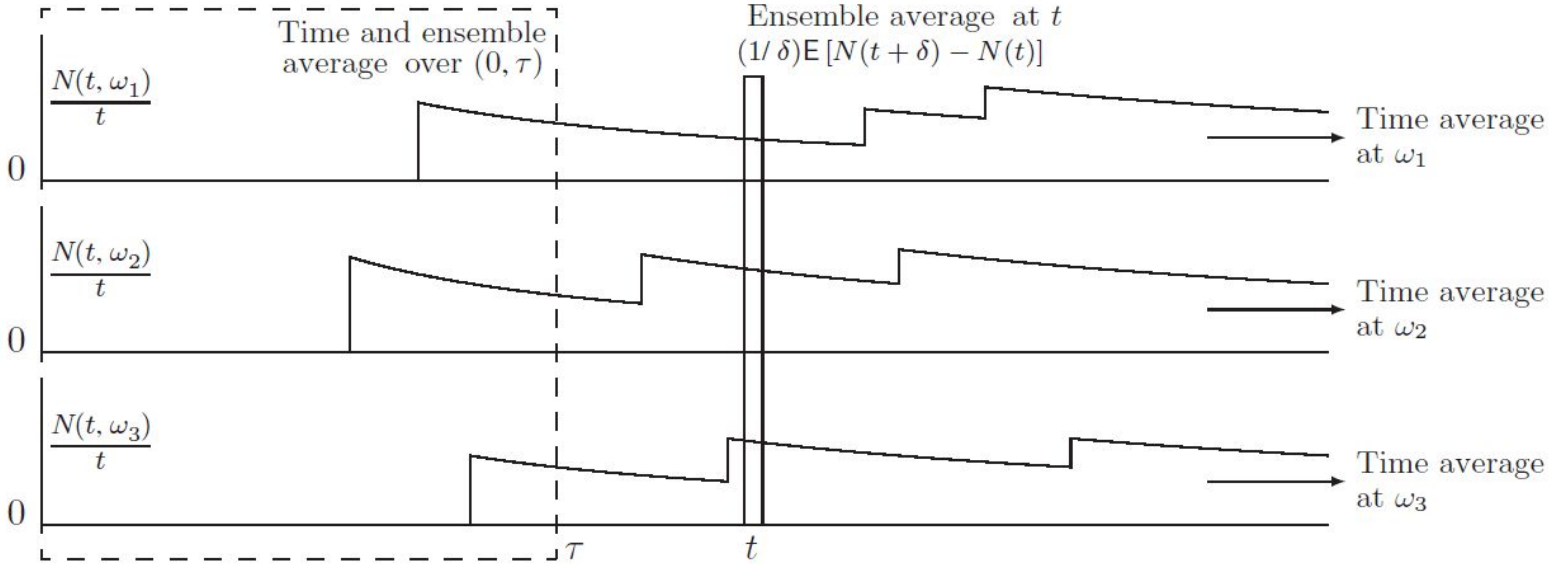
$$\lim_{t\to\infty} \frac{N(t)}{S_{N(t)+1}} = \lim_{n\to\infty} \frac{n}{S_{n+1}} = \lim_{n\to\infty} \frac{n+1}{S_{n+1}} \frac{n}{n+1} = \frac{1}{\overline{X}} \quad \text{WP1.}$$

Since $N(t)/t$ is between these two quantities with the same limit,

$$\lim_{t\to\infty} N(t)/t = 1/\overline{X} \quad \text{WP1.}$$

§ 5.3 Strong law for renewal processes

The time average at a sample point ω, the time and ensemble average from 0 to a given τ, and the ensemble average in an interval $(t, t+\delta]$.



This equality of time averages and limiting ensemble averages for renewal processes carries over to a large number of stochastic processes, and forms the basis of ergodic theory.

§ 5.3 Strong law for renewal processes

Note that in order to equate time averages and limiting ensemble averages, quite a few conditions are required.

- First, the time average must exist in the limit $t \to \infty$ with probability 1; that time average must also be the same for all sample points in a set of probability 1.
- Second, the ensemble average must approach a limit as $t \to \infty$.
- Third, the time average and ensemble average must be the same.

§ 5.3 Strong law for renewal processes

Counter example

Let $\{X_i; i \geqslant 1\}$ be a sequence of binary i.i.d. rv.s, each taking the value 0 with probability $1/2$ and 2 with probability $1/2$. Let $M_n = X_1 X_2 \cdots X_n$.

- We see that $\lim_{n\to\infty} M_n = 0$ WP1.
- Also $\mathrm{E}[M_n] = 1$ for all $n \geqslant 1$.

Thus the time average exists and equals 0 WP1 and the ensemble average exists and equals 1 for all n, but the two are different.

§ 5.3 Strong law for renewal processes

Theorem 3 (CLT for $N(t)$)

Assume that the inter-renewal intervals for a renewal counting process $\{N(t); t > 0\}$ have finite standard deviation $\sigma > 0$. Then

$$\lim_{t\to\infty} P\left\{\frac{N(t) - t/\overline{X}}{\sigma\overline{X}^{-3/2}\sqrt{t}} < \alpha\right\} = \Phi(\alpha),$$

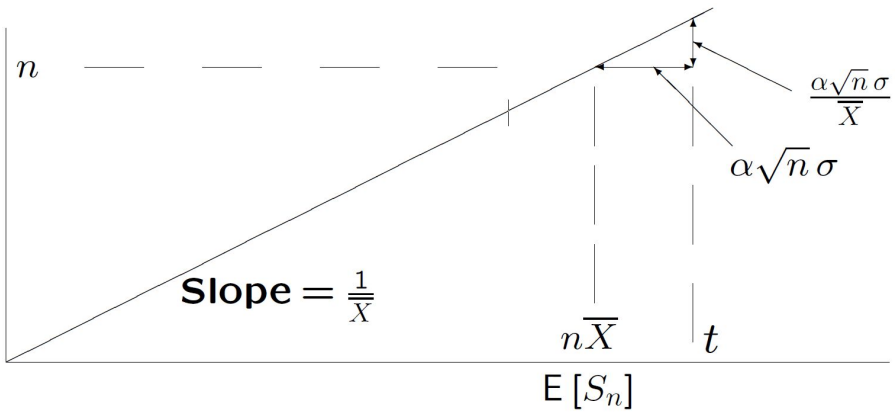
where

$$\Phi(y) = \int_{-\infty}^{y} \frac{1}{\sqrt{2\pi}} \exp(-x^2/2)\mathrm{d}x.$$

This says that the cumulative distribution function (CDF) of $N(t)$ tends to the Gaussian distribution with mean $t/\overline{X}$ and standard deviation $\sigma\overline{X}^{-3/2}\sqrt{t}$.

§ 5.3 Strong law for renewal processes

Idea of proof: For t sufficiently large,



S_n 的均值是 $n\bar{X}$, 方差是 $n\sigma^2$, 所以，$\frac{S_n - n\bar{X}}{\sqrt{n}\sigma}$ 可以用标准正态分布近似描述。$\{S_n \le t\}$ 等价于 $\{N(t) \ge n\}$.

$$t = n\bar{X} + \alpha\sqrt{n}\sigma$$
$$n \approx t/\bar{X} - \alpha\sigma\sqrt{t/\bar{X}}/\bar{X}$$

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Time-average residual life

Time-average age

Time-average duration

General Renewal-reward Processes

§ 5.4 Renewal–reward processes: time-average residual life

Definition 7

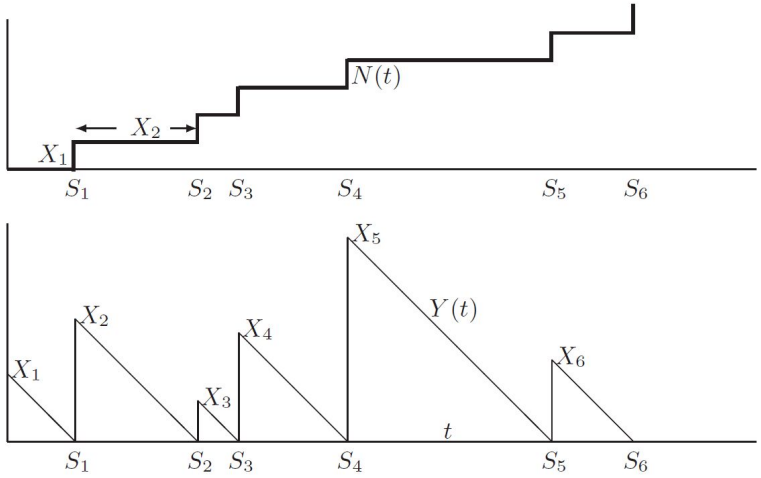
For a renewal process $\{N(t); t > 0\}$, the residual life $Y(t)$ at time t is the remaining time until the next renewal, i.e., $Y(t) = S_{N(t)+1} - t$.

It's how long you have to wait for a bus (if bus arrivals were renewal processes).

We can view residual life as a reward function on a renewal process. The sample reward at t is a function of the sample path of renewals.

The residual life, as a function of t, is a random process, and we can look at its time-average value, $[\int_0^t Y(\tau)\mathrm{d}\tau]/t$.

§ 5.4 Renewal–reward processes: time-average residual life

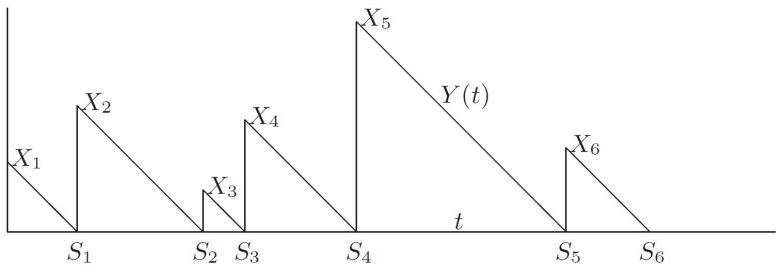


Note that a residual-life sample function is a sequence of isosceles right triangles, with part of a final triangle at the end. Thus

$$\int_0^t y(\tau)\mathrm{d}\tau = \frac{1}{2}\sum_{i=1}^{n(t)} x_i^2 + \int_{\tau=s_{n(t)}}^{t} y(\tau)\mathrm{d}\tau,$$

where $\{x_i; 0 < i < \infty\}$ is the set of sample values for the inter-renewal intervals.

§ 5.4 Renewal-reward processes: time-average residual life



$$\frac{1}{2t}\sum_{n=1}^{N(t)} X_n^2 \leqslant \frac{1}{t}\int_0^t Y(\tau)\mathrm{d}\tau \leqslant \frac{1}{2t}\sum_{n=1}^{N(t)+1} X_n^2.$$

$$\lim_{t\to\infty}\frac{\sum_{n=1}^{N(t)} X_n^2}{2t} = \lim_{t\to\infty}\frac{\sum_{n=1}^{N(t)} X_n^2}{N(t)}\frac{N(t)}{2t} = \frac{\mathrm{E}[X^2]}{2\mathrm{E}[X]} \qquad \text{WP1.}$$

$$\lim_{t\to\infty}\frac{\sum_{n=1}^{N(t)+1} X_n^2}{2t} = \lim_{t\to\infty}\frac{\sum_{n=1}^{N(t)+1} X_n^2}{N(t)+1}\frac{N(t)+1}{N(t)}\frac{N(t)}{2t} = \frac{\mathrm{E}[X^2]}{2\mathrm{E}[X]} \qquad \text{WP1.}$$

$$\lim_{t\to\infty}\frac{\int_0^t Y(\tau)\mathrm{d}\tau}{t} = \frac{\mathrm{E}[X^2]}{2\mathrm{E}[X]} \qquad \text{WP1.}$$

§ 5.4 Renewal–reward processes: time-average residual life

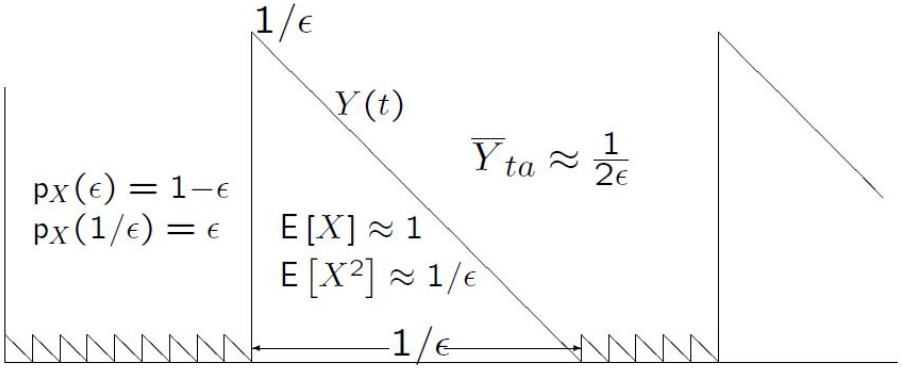
Time average residual life $\overline{Y}_{ta} = \frac{\mathrm{E}[X^2]}{2\mathrm{E}[X]}$.

If X is exponential (w.r.t. Poisson process), $\overline{Y}_{ta} = \mathrm{E}[X]$.

If X is almost deterministic, $\overline{Y}_{ta} \approx \frac{\mathrm{E}[X]}{2}$.

Example 1

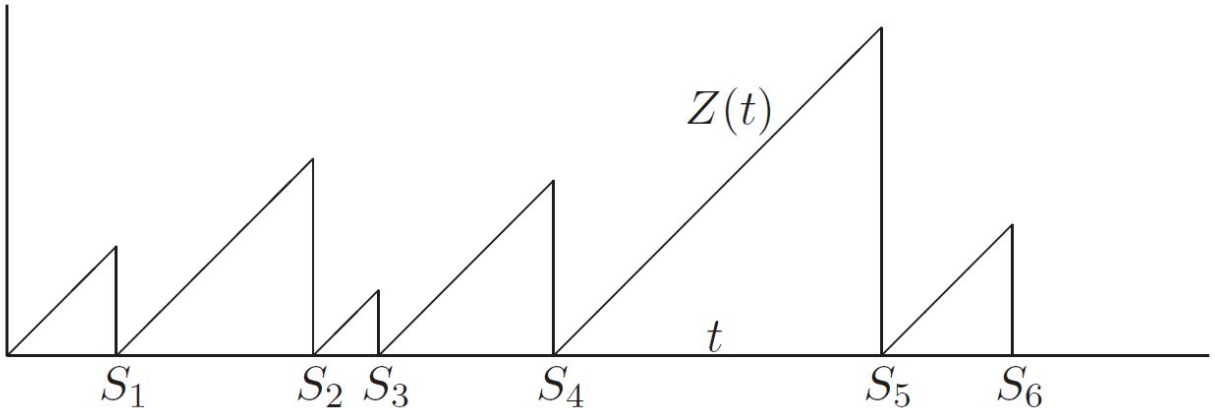
As an example of the effect of improbable but large inter-renewal intervals, let X take on the value ϵ with probability $1-\epsilon$ and value $1/\epsilon$ with probability ϵ.



§ 5.4 Renewal–reward processes: time-average age

Definition 8

Let $Z(t)$ be the age of a renewal process at time t where age is defined as the interval from the most recent arrival before (or at) t until t, i.e., $Z(t) = t - S_{N(t)}$.



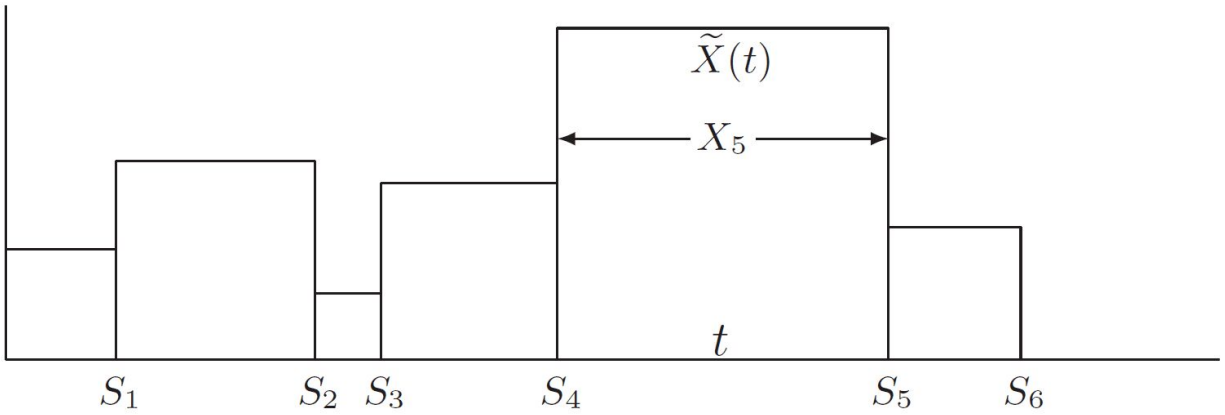
$$\lim_{t\to\infty} \frac{\int_0^t Z(\tau)\mathrm{d}\tau}{t} = \frac{\mathrm{E}[X^2]}{2\mathrm{E}[X]} \qquad \text{WP1.}$$

§ 5.4 Renewal–reward processes: time-average duration

Definition 9

Let $\widetilde{X}(t)$ be the duration of the inter-renewal interval containing time t, i.e., $\widetilde{X}(t) = X_{N(t)+1} = S_{N(t)+1} - S_{N(t)}$.

It is clear that $\widetilde{X}(t) = Y(t) + Z(t)$.



$$\lim_{t\to\infty} \frac{\int_0^t \widetilde{X}(\tau)\mathrm{d}\tau}{t} = \frac{\mathrm{E}[X^2]}{\mathrm{E}[X]} \qquad \text{WP1.}$$

5.4.1 General renewal-reward processes

We now investigate the general class of reward functions for which the reward at time t depends at most on the age and the duration at t, i.e., the reward $R(t)$ at time t is given explicitly as a function $R(Z(t), \widetilde{X}(t))$ of the age and duration at t.

We want to find the time-average value of $R(t)$, namely, $\lim_{t\to\infty}[\int_0^t R(\tau)\mathrm{d}\tau]/t$.

We first define R_n as the accumulated reward in the nth renewal interval,

$$\begin{aligned} R_n \triangleq \int_{S_{n-1}}^{S_n} R(\tau)\mathrm{d}\tau &= \int_{S_{n-1}}^{S_n} R(Z(\tau), \widetilde{X}(\tau))\mathrm{d}\tau \\ &= \int_{S_{n-1}}^{S_n} R(\tau - S_{n-1}, X_n)\mathrm{d}\tau \\ &= \int_{z=0}^{X_n} R(z, X_n)\mathrm{d}z. \end{aligned}$$

This is a function only of the r.v. X_n.

5.4.1 General renewal-reward processes

In general, the expected value of R_n is given by

$$\mathrm{E}[R_n] = \int_{x=0}^{\infty} \left(\int_{z=0}^{x} R(z,x)\mathrm{d}z \right) \mathrm{d}F_X(x).$$

Theorem 1

Let $\{R(t); t > 0\}$ be a non-negative renewal-reward function for a renewal process with expected inter-renewal time $\mathrm{E}[X] = \overline{X} < \infty$. If each R_n is a r.v. with $\mathrm{E}[R_n] < \infty$, then

$$\lim_{t\to\infty} \frac{1}{t} \int_{\tau=0}^{t} R(\tau)\mathrm{d}\tau = \frac{\mathrm{E}[R_n]}{\overline{X}} \quad \text{WP1}.$$

Proof:

$$\frac{\sum_{n=1}^{N(t)} R_n}{t} \leqslant \frac{\int_0^t R(\tau)\mathrm{d}\tau}{t} \leqslant \frac{\sum_{n=1}^{N(t)+1} R_n}{t}.$$

5.4.1 General renewal-reward processes

$$\frac{\sum_{n=1}^{N(t)} R_n}{t} = \frac{\sum_{n=1}^{N(t)} R_n}{N(t)} \frac{N(t)}{t} \to \frac{\mathrm{E}[R_n]}{\overline{X}} \quad \text{WP1.}$$

$$\frac{\sum_{n=1}^{N(t)+1} R_n}{t} = \frac{\sum_{n=1}^{N(t)+1} R_n}{N(t)+1} \frac{N(t)+1}{N(t)} \frac{N(t)}{t} \to \frac{\mathrm{E}[R_n]}{\overline{X}} \quad \text{WP1.}$$

The same result holds for non-positive reward functions.

Corollary 1

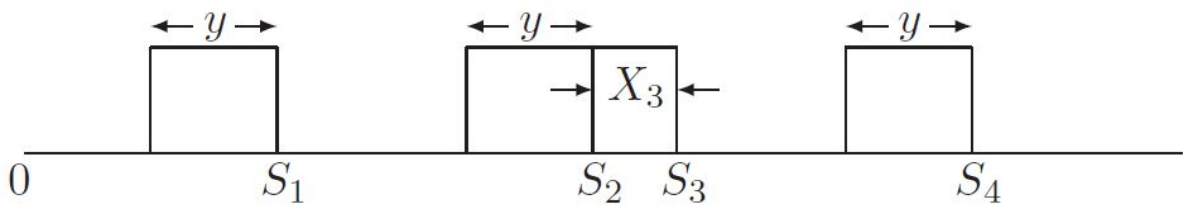
Let $\{R(t); t > 0\}$ be a renewal-reward function for a renewal process with expected inter-renewal time $\mathrm{E}[X] = \overline{X} < \infty$. If each R_n is a rv. with $\mathrm{E}[|R_n|] < \infty$, then

$$\lim_{t\to\infty} \frac{1}{t} \int_{\tau=0}^{t} R(\tau)\mathrm{d}\tau = \frac{\mathrm{E}[R_n]}{\overline{X}} \quad \text{WP1.}$$

5.4.1 General renewal-reward processes

Example 2: Distribution of residual life

We would like to find the time average CDF of $Y(t)$, i.e., the fraction of time that $Y(t) \leqslant y$ for any given y.



$$R(t) = \begin{cases} 1, & \widetilde{X}(t) - Z(t) \leqslant y, \\ 0, & \text{otherwise.} \end{cases}$$

Thus, $R_n = \min\{y, X_n\}$.

$$\mathrm{E}[R_n] = \mathrm{E}[\min\{X, y\}] = \int_{x=0}^{y} \Pr\{X > x\} \mathrm{d}x.$$

$$F_Y(y) = \frac{\mathrm{E}[R_n]}{\overline{X}} = \frac{1}{\overline{X}} \int_{x=0}^{y} \Pr\{X > x\} \mathrm{d}x \quad \text{WP1.}$$

课后作业

作业 1. 以下是否正确：

(1) 是否 $N(t) < n$ 当且仅当 $S_n > t$?

(2) 是否 $N(t) \leq n$ 当且仅当 $S_n \geq t$?

(3) 是否 $N(t) > n$ 当且仅当 $S_n < t$?

作业 2： 假设更新过程的到达间隔分布是均值为 μ 的泊松分布。即假设

$$\mathrm{P}\{\mathrm{X_n} = \mathrm{k}\} = \mathrm{e}^{-\mu}\frac{\mu^{\mathrm{k}}}{\mathrm{k!}},\ \mathrm{k} = 0, 1, \cdots$$

(1) 求 S_n 的分布.

(2) 计算 $\mathrm{P}\{\mathrm{N(t)} = \mathrm{n}\}$.

课后作业

作业 3： 设一个更新过程，间隔 X 服从分布 $\Pr\{X=1\}=1/2$, $\Pr\{X=2\}=1/2$。画出 5 条计数样本 $N(t), t \le 5$. 试着讨论 S_3 的分布律，$N(3)$ 的分布律。

作业 4： 令 $\{N_1(t), t \ge 0\}$ 和 $\{N_2(t), t \ge 0\}$ 是独立的更新过程。令 $N(t) = N_1(t) + N_2(t)$.

(1) $\{N(t), t \ge 0\}$ 的到达间隔时间是否独立？

(2) 它们是否同分布？

(3) $\{N(t), t \ge 0\}$ 是否是更新过程？

课后作业

作业 5： 以 S_n 记具有到达间隔分布函数 F 的更新过程 $\{N(t), t \geq 0\}$ 的第 n 个事件时刻。

(1) 问 $P\{N(t) = n | S_n = y\}$ 是什么？

(2) 从 $P\{N(t) = n\} = \int_0^\infty P\{N(\mathrm{t}) = n | S_n = y\} f_{S_n}(y)\,\mathrm{d}y$ 开始，利用 n 个速率为 λ 的独立指数随机变量的和具有参数为 (n, λ) 的伽马分布，当 $F(y) = 1 - e^{-\lambda y}$ 时计算 $P\{N(t) = n\}$。

课后作业

作业 6： 考虑一个平均到达间隔时间为 μ 的更新过程。假设这个过程的每一个事件以概率 p 被记入。以 $N_C(t)$ 计到时刻 $t(t>0)$ 为止被计入的事件数。

(1)$\{N_C(t), t \geq 0\}$ 是更新过程吗？

(2)$\lim_{t\to\infty} \frac{N_C(t)}{t}$ 是多少？

作业 7： 令 $U_1, U_2, \ldots$ 是独立的 (0,1) 均匀随机变量，定义 N 为 $N = \min\{n : U_1 + U_2 + \cdots + U_n > 1\}$，$E[N]$ 是多少？

课后作业

作业 8： Let $\{N(t); t > 0\}$ be a renewal counting process with inter-renewal rv.s $\{X_n; n \geqslant 1\}$. Then (whether or not $\overline{X} < \infty$), $\lim_{t\to\infty} N(t) = \infty$ WP1 and $\lim_{t\to\infty} \mathrm{E}[N(t)] = \infty$.

（1）请论证每个样本函数都有极限，或有限，或无穷。（因为样本函数是不减的）

（2）请论证 $\{N(t) < n\}$ 这个事件应包含事件 $\{\lim_{t\to\infty} N(t) < n\}$ 所以，$\Pr\{\lim N(t) < n\} \leq P\{N(t) < n\} = P\{S_n > t\} \to 0$, 因为 S_n 是随机变量。

（3）证明 $\lim_{t\to\infty} \mathrm{E}[N(t)] = \infty$。

课后作业

作业 9： 假设乘客按照一个更新过程来到一火车站，其平均来到间隔时间为 μ。每当有 N 个人在车站上等待时，就开出一辆火车。若每当有 n 个乘客等待时，车站就以每单位时间 nc 元的比率开支费用，且每开出一辆火车要多开支 K 元，那么此车站每单位时间的平均费用是多少？

作业 10： 科学研究是磨练心智的过程。一个研究生的情绪或有周期现象。周期是两天或四天的机会均等且互不影响。设一个周期内快乐指数 h 是以区间中点为圆心的半圆曲线。

(1) 画图示意一个快乐指数的样本。

(2) 若一个研究生快乐指数大于 3/4，我们称他是快乐的。试求长时间观察，一个研究生平均快乐的时间的比例。

Thank you for your attention!